

QMS 101 Introductory Statistics

Topic III: Measures of Central Tendency

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Learning Outcomes

By the end of this topic, students will be able to:

- ▶ Explain the meaning and purpose of a measure of central tendency
- ▶ Calculate the **arithmetic mean** for ungrouped and grouped data
- ▶ Find the **median** and **mode** for ungrouped and grouped data
- ▶ Compute the **geometric mean** and **harmonic mean** and know when to use each
- ▶ Find **quartiles**, **deciles**, and **percentiles** from a dataset
- ▶ Choose the most appropriate measure for a given situation and **interpret** every result in plain language

Our Running Dataset — Used Throughout This Topic

The Study Context

A farm manager in a rural village wants to estimate the typical monthly expenditure on farm inputs (such as seeds, fertilizers, and pesticides) among small-scale farmers.

The raw data she collected (**TZS '000**) from 30 farmers were:

45	32	55	38	47	60	33	50	42	38
55	47	30	65	42	50	38	55	47	44
52	36	48	58	41	53	37	61	43	49

We will use the following frequency distribution:

Class (TZS '000)	Class boundary	f	Midpoint x	F
30 – 38	29.5 - 38.5	7	34	7
39 – 47	38.5 - 47.5	10	43	17
48 – 56	47.5 - 56.5	7	52	24
57 – 65	56.5 - 65.5	6	61	30
Total		30		



Tip

Every measure in this topic will be calculated from this same dataset.

You will see how each measure tells a different part of the same story.

3.1 What is a Measure of Central Tendency?

The Question Every Decision-Maker Asks

Real-Life Scenario

A farm manager in a rural village wants to estimate the typical monthly expenditure on farm inputs (such as seeds, fertilizers, and pesticides) among small-scale farmers.

- ▶ She has collected spending data from 30 farmers.
- ▶ However, she cannot present all 30 values to the finance officer.
- ▶ Instead, she needs one representative value that best describes a typical farmer's monthly spending.
*This single representative value is called a **measure of central tendency**.*

The Question Every Decision-Maker Asks

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- ▶ However, she cannot present all 30 values to the finance officer.
- ▶ Instead, she needs one representative value that best describes a typical farmer's monthly spending.
*This single representative value is called a **measure of central tendency**.*

Definition

A **measure of central tendency** is a single numerical value that represents the *centre* or *typical value* of a dataset. It summarises an entire distribution into one meaningful number.

Five Measures — One Concept, Five Answers

The same question — “*What is the typical monthly spending?*” — can be answered five different ways:

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Measure	What It Finds	Best Situation
Arithmetic Mean	Balancing point of all values	Symmetric data, no extreme outliers
Median	The exact middle value	Skewed data or when outliers are present
Mode	The most common value	Categorical data; finding the most popular outcome
Geometric Mean	Average of growth rates or ratios	Growth rates, index numbers, ratios
Harmonic Mean	Average of rates	Speed, price per unit, productivity rates

Five Measures — One Concept, Five Answers

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 Warning

No single measure is always correct. Part of learning statistics is knowing **which measure to choose** — and being able to **explain why**.

3.2 The Arithmetic Mean

Definition and Formula

The **arithmetic mean** ($\bar{x}$) is calculated by adding all values and dividing by the number of observations.

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Ungrouped data:

$$\bar{x} = \frac{\sum x}{n}$$

$\sum x$ = sum of all values

n = number of observations

Grouped data (frequency table):

$$\bar{x} = \frac{\sum fx}{\sum f}$$

f = class frequency

x = class **midpoint** (not limits!)

$$\sum f = n$$

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$$\sum f = n$$



Warning

Critical rule: For grouped data, always use the class **midpoint** as x .

The midpoint = $\frac{\text{lower limit} + \text{upper limit}}{2}$

Never use the class limits themselves.

Arithmetic Mean — Ungrouped (from raw data)

Using the 30 raw values of monthly farm input expenditure (TZS '000):

$$\begin{aligned}\sum x = & 45 + 32 + 55 + 38 + 47 + 60 + 33 + 50 + 42 + 38 + \\ & 55 + 47 + 30 + 65 + 42 + 50 + 38 + 55 + 47 + 44 + 52 + 36 + \\ & 48 + 58 + 41 + 53 + 37 + 61 + 43 + 49 \\ & \dots\end{aligned}$$

$$\bar{x} = \frac{\sum x}{n} = \frac{1,391}{30} = \mathbf{TZS\ 46,367}$$

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$$\bar{x} = \frac{\sum x}{n} = \frac{1,391}{30} = \mathbf{TZS\ 46,367}$$

Interpretation

- ▶ The average monthly expenditure on farm inputs among the 30 farmers is approximately **TZS 46,367**.
- ▶ This means that a typical small-scale farmer spends about this amount per day on agricultural inputs such as seeds, fertilizers, and pesticides.
- ▶ A farm manager or policy planner could use this value as a benchmark for planning input subsidies or estimating average production costs in the area.

Arithmetic Mean — Grouped Data

Using the frequency distribution:

Class	f	Midpoint x	fx
30 – 38	7	34	238
39 – 47	10	43	430
48 – 56	7	52	364
57 – 65	6	61	366
Total	30		$\sum fx = 1,398$

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$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1,398}{30} = \textbf{TZS 46,600}$$

Arithmetic Mean — Grouped Data

Using the frequency distribution:

Class	f	Midpoint x	fx
30 – 38	7	34	238
39 – 47	10	43	430
48 – 56	7	52	364
57 – 65	6	61	366
Total	30		$\sum fx = 1,398$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1,398}{30} = \textbf{TZS 46,600}$$

Why the slight difference? (The ungrouped mean = TZS 46,367. The grouped mean = TZS 46,600.)

Because we assumed every observation in a class sits at the **midpoint** — an approximation. The two values are very close.

Practice Question 3.1 — Arithmetic Mean

Try This

Q1 A student recorded the number of **hours slept per night** for 10 classmates:

7, 6, 8, 5, 7, 9, 6, 7, 8, 6

- (a) Calculate the arithmetic mean number of hours slept.
- (b) The recommended sleep for a university student is 8 hours per night.

By how much does the average student in this group fall short of the recommendation?

Q2 Using the grouped data on small-scale farmers' monthly farm input expenditure, the mean is about TZS 46.6 thousand.

- (a) A farmer spends TZS 30,000 per day. How does this compare to the average, and what does it suggest?
- (b) A farmer spends TZS 60,000 per day. How does this compare to the average, and what does it suggest?

3.3 The Median

Definition

Definition

The **median** is the *middle value* when data is arranged in ascending order.

- ▶ Exactly **50%** of values lie below the median
- ▶ Exactly **50%** of values lie above the median

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For ungrouped data:

Situation	Position of median
n is odd	$\frac{n+1}{2}$ th value
n is even	Average of $\frac{n}{2}$ th and $\left(\frac{n}{2} + 1\right)$ th values

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Warning

Why the Median Matters — Outlier Resistance

A Quick Illustration

Eight farmers spend (TZS '000/month): **200, 250, 280, 300, 320, 350, 400, 3,500**

Mean = $5,600 / 8 = \text{TZS } 700,000$

Median = average of 4th and 5th values = $(300 + 320)/2 = \text{TZS } 310,000$

Interpretation - Most farmers spend between TZS 200,000 and 400,000. One extreme value (TZS 3,500,000) is an outlier. This outlier pulls the *mean upward* (TZS 700,000) making it unrepresentative of a typical farmer.

- ▶ The median (TZS 310,000) better represents the typical spending level, since it is not affected by extreme values.

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- ▶ The median (TZS 310,000) better represents the typical spending level, since it is not affected by extreme values.

This is why **income data, house prices, and waiting times** are almost always reported using the **median** — not the mean.

Median from Ungrouped Data — Our Dataset

Sorting all 30 spending values in ascending order:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

Median from Ungrouped Data — Our Dataset

Sorting all 30 spending values in ascending order:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

$n = 30$ (even) $\rightarrow$ average of **15th** and **16th** values

15th value = 47, 16th value = 47

$$M_d = \frac{47 + 47}{2} = \textbf{TZS 47,000}$$

Median from Ungrouped Data — Our Dataset

Sorting all 30 spending values in ascending order:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

$n = 30$ (even) $\rightarrow$ average of **15th** and **16th** values

15th value = 47, 16th value = 47

$$M_d = \frac{47 + 47}{2} = \textbf{TZS 47,000}$$

Interpretation

Half of the small-scale farmers spend less than TZS 47,000 per day on farm inputs, and the **other half** spend more than TZS 47,000.

Median from Grouped Data

When data is in a frequency table, we estimate the median using:

$$M_d = L + \left[\frac{\frac{n}{2} - F}{f} \right] \times h$$

Median from Grouped Data

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$$M_d = L + \left[\frac{\frac{n}{2} - F}{f} \right] \times h$$

Symbol	Meaning
L	Lower boundary of the median class
n	Total observations
F	Cumulative frequency of all classes before the median class
f	Frequency of the median class
h	Class width
Median class	The class where cumulative frequency first reaches $n/2$

Grouped Median — Step by Step

Class (TZS '000)	Class boundary	f	F
30 – 38	29.5 - 38.5	7	7
39 – 47	38.5 - 47.5	10	17
48 – 56	47.5 - 56.5	7	24
57 – 65	56.5 - 65.5	6	30
Total		30	

Grouped Median — Step by Step

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57 – 65	56.5 – 65.5	6	30
Total		30	

Step 1: $n/2 = 30/2 = 15 \rightarrow$ median class is **39–47**

Step 2: Identify components: $L = 38.5$, $F = 7$, $f = 10$, $h = 9$

. . . .

$$M_d = L + \left[\frac{\frac{n}{2} - F}{f} \right] \times h = 38.5 + \left[\frac{15 - 7}{10} \right] \times 9 = \textbf{TZS 45,700}$$

. . .

$$M_d = L + \left[\frac{\frac{n}{2} - F}{f} \right] \times h = 38.5 + \left[\frac{15 - 7}{10} \right] \times 9 = \textbf{TZS 45,700}$$

Note on the Difference

Ungrouped median = TZS 47,000 | Grouped median = TZS 45,700

The grouped formula is an **estimate** based on the assumption that data is spread evenly within each class.

Practice Question 3.2 — Median

Try This

The table below shows the **monthly mobile data usage (GB)** of 25 students:

Class (GB)	Frequency f
1 – 5	4
6 – 10	7
11 – 15	8
16 – 20	4
21 – 25	2

- (a) Build the Class boundary and cumulative frequency columns.
- (b) Identify the median class. Show how you found it.
- (c) Apply the interpolation formula to estimate the median data usage.

3.4 The Mode

Definition and Types

Definition

The **mode** is the value (or class) that appears **most frequently** in a dataset.

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Type	Meaning
Unimodal	One mode
Bimodal	Two modes
Multimodal	More than two
No mode	All values equal frequency

Definition and Types

Definition

The **mode** is the value (or class) that appears **most frequently** in a dataset.

Type	Meaning	Unique Advantage
Unimodal	One mode	The mode is the only central tendency measure that works for qualitative (categorical) data. Example: <i>Most popular mobile network among students = M-Pesa (22 out of 50)</i> You cannot compute a mean or median of network names — but you can find the most frequent one.
Bimodal	Two modes	
Multimodal	More than two	
No mode	All values equal frequency	

Mode from Ungrouped Data — Our Dataset

Sorted data:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

Mode from Ungrouped Data — Our Dataset

Sorted data:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

Frequency count of repeated values:

Value	38	42	47	50	55
Count	3	2	3	2	3

Mode from Ungrouped Data — Our Dataset

Sorted data:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

Frequency count of repeated values:

Value	38	42	47	50	55
Count	3	2	3	2	3

Values 38, 47, and 55 each appear **3 times** → **Multi-modal:** $M_o =$
TZS 38,000, 47,000, and 55,000

Mode from Ungrouped Data — Our Dataset

Sorted data:

30	32	33	36	37	38	38	38	41	42
42	43	44	45	47	47	47	48	49	50
50	52	53	55	55	55	58	60	61	65

Frequency count of repeated values:

Value	38	42	47	50	55
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Values 38, 47, and 55 each appear **3 times** → **Multi-modal:** $M_o =$
TZS 38,000, 47,000, and 55,000

Interpretation

The modes are TZS 38,000, 47,000, and 55,000 because these spending values occur most frequently among the farmers. This means these are the most common monthly spending levels on

Mode from Grouped Data — Formula

For grouped data, the mode is **estimated** from the modal class:

$$M_o = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

Mode from Grouped Data — Formula

For grouped data, the mode is **estimated** from the modal class:

$$M_o = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

Sym- bol	Meaning
-------------	---------

L	Lower boundary of the modal class (class with highest f)
-----	---

f_1	Frequency of the modal class
-------	------------------------------

f_0	Frequency of the class immediately before the modal class
-------	--

f_2	Frequency of the class immediately after the modal class
-------	---

h	Class width
-----	-------------

Grouped Mode — Step by Step

Class (TZS '000)	Class boundary	f
30 – 38	29.5 – 38.5	7
39 – 47	38.5 – 47.5	10
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Total		30

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Total		30

Identify components:

$$L = 38.5, f_0 = 7, f_1 = 10, f_2 = 7, h = 9$$

Grouped Mode — Step by Step

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30 – 38	29.5 - 38.5	7
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Total		30

Identify components:

$$L = 38.5, f_0 = 7, f_1 = 10, f_2 = 7, h = 9$$

$$M_o = 38.5 + \left[\frac{10 - 7}{2(10) - 7 - 7} \right] \times 9 = 38.5 + \left[\frac{3}{6} \right] \times 9 = \textbf{TZS 43,000}$$

Grouped Mode — Step by Step

Class (TZS '000)	Class boundary	<i>f</i>
30 – 38	29.5 – 38.5	7
39 – 47	38.5 – 47.5	10
48 – 56	47.5 – 56.5	7
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Interpretation

The most common daily farm input expenditure among the small scale farmers is approximately TZS 43,000

Skewness: What the Order of Measures Tells You

Symmetric (No skew)

Right-skewed

Left-skewed

Mean = Median =
Mode

→
Mean > Median >
Mode
Our dataset fall here

→
Mean < Median <
Mode

Summary: from the Example

Mean = TZS 46,600 | Median = TZS 45,700 | Mode = TZS 43,000

The order Mean > Median > Mode indicates a **right skew** — a few high spenders are pulling the mean upward.

Practice Question 3.3 — Mode

Try This

The following frequency table shows the **daily number of customer complaints** received by a shop over 30 days:

Class (complaints)	Frequency f
0 – 2	5
3 – 5	12
6 – 8	8
9 – 11	4
12 – 14	1

- (a) Identify the modal class.
- (b) Calculate the estimated mode using the interpolation formula.
- (c) Given that the mean number of monthly complaints is 5.3, what does the relationship between mean and mode tell you

3.5 The Geometric Mean

Why Do We Need Another Type of Mean?

Note

In our study of small-scale farmers' daily expenditure on farm inputs, we used:

- ▶ Arithmetic mean → average spending
 - ▶ Median → middle spending
 - ▶ Mode → most common spending level
-
- ▶ These measures are appropriate because expenditure is additive data (values are summed).
 - ▶ Why the Arithmetic Mean Fails for Growth Data

Example

A student invests TZS 1,000,000:

- ▶ Year 1: grows **+100%** → TZS 2,000,000
- ▶ Year 2: falls **−50%** → back to TZS 1,000,000
- ▶ **Arithmetic mean return** =
$$\frac{100 + (-50)}{2} = \mathbf{25\% \text{ per year}}$$
- ▶ In reality: But the student's money did **not grow at all!**
The arithmetic mean **overstates** the return because growth rates *compound* — each year's gain or loss acts on the previous year's total, not on the original amount.

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The arithmetic mean **overstates** the return because growth rates *compound* — each year's gain or loss acts on the previous year's total, not on the original amount.

The **geometric mean** is the correct tool for averaging growth rates, ratios, and any multiplicative process.

Definition and Formula

Definition

The **geometric mean** is the n th root of the product of n positive values.

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Direct formula:

$$GM = \sqrt[n]{x_1 \times x_2 \times \cdots \times x_n}$$

Log formula (easier for large n):

$$GM = \text{Antilog} \left(\frac{\sum \log x}{n} \right)$$

Here “*Antilog*” is 10^y where
 $y = \left(\frac{\sum \log x}{n} \right)$

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Here “*Antilog*” is 10^y where
 $y = \left(\frac{\sum \log x}{n}\right)$

! Important

All values must be positive. The GM cannot be computed if any value is zero or negative.

This is why we convert percentage rates like -5% to a **growth factor** of 0.95 before computing the GM.

When to Use the Geometric Mean

Use GM when data involves:

- ▶ **Growth rates** (GDP growth, investment returns, population growth)
- ▶ **Percentage changes** (price indices, inflation rates)
- ▶ **Ratios** (exchange rates, financial ratios)
- ▶ **Data spanning very wide ranges** (earthquake magnitudes, population sizes)

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Tip

Simple test:

If the values *multiply together* to give a meaningful result → use GM.

If the values *add together* to give a meaningful result → use arithmetic mean.

Connecting GM to Our Dataset

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Warning

Our dataset measures **monthly spending amounts in TZS** — not growth rates or ratios. So GM here is not appropriate. We introduce GM now with a separate example

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Appropriate GM Example — Investment Returns

A finance student's portfolio grew at these annual rates over 5 years:

8%, 12%, -4%, 15%, 6%

Convert to growth factors: 1.08, 1.12, 0.96, 1.15, 1.06

Formula

Geometric Mean — Solved Step by Step

Data: 1.08, 1.12, 0.96, 1.15, 1.06

Geometric Mean — Solved Step by Step

Data: 1.08, 1.12, 0.96, 1.15, 1.06

Using the log formula:

x (growth factor)	$\log_{10} x$
1.08	0.03342
1.12	0.04922
0.96	-0.01773
1.15	0.06070
1.06	0.02531
Sum	$\sum \log x = 0.15092$

Geometric Mean — Solved Step by Step

Data: 1.08, 1.12, 0.96, 1.15, 1.06

Using the log formula:

x (growth factor)	$\log_{10} x$
1.08	0.03342
1.12	0.04922
0.96	-0.01773
1.15	0.06070
1.06	0.02531
Sum	$\sum \log x = 0.15092$

$$\log(GM) = \frac{0.15092}{5} = 0.030184$$

$$GM = \text{Antilog}(0.030184) = 10^{0.030184} = \mathbf{1.0718}$$

Verifying the Geometric Mean

Verify!

If the portfolio grows at a **constant** 7.18% per year for 5 years:

$$\text{TZS } 1,000,000 \times (1.0718)^5 = \text{TZS } 1,000,000 \times 1.4148 = \text{TZS } 1,414,800$$

Now check using the actual rates:

$$1,000,000 \times 1.08 \times 1.12 \times 0.96 \times 1.15 \times 1.06 = \text{TZS } 1,414,800 \checkmark$$

Arithmetic mean would give $(8 + 12 - 4 + 15 + 6)/5 = 7.4\%$ per year

→ $\text{TZS } 1,000,000 \times 1.074^5 = \text{TZS } 1,429,000$ — **an overestimate**. The GM gives the exact correct answer.

Practice Questions — Geometric Mean

Q1

The Consumer Price Index (CPI) in a country changed by the following rates over four years:

Year	1	2	3	4
Rate	+6%	+9%	+3%	+12%

- (a) Convert each rate to a growth factor.
- (b) Calculate the geometric mean annual inflation rate using the log formula.
- (c) If a basket of goods cost TZS 50,000 at the start of Year 1, what would it cost at the end of Year 4?
- (d) Why would using the arithmetic mean give an incorrect answer here?

Q2

A university library is tracking how many students are using their new “E-Book Portal.” The usage grew by the following annual rates over 3 years:

- ▶ Year 1: +20% growth (many early adopters)
- ▶ Year 2: +40% growth (marketing campaign success)
- ▶ Year 3: +5% growth (market saturation)

Calculate the Geometric mean using direct formular